

# Chapter 1, Section 4

James Lee

October 9, 2026

1. If  $f$  and  $g$  are regular functions on open subsets  $U$  and  $V$  of a variety  $X$ , and if  $f = g$  on  $U \cap V$ , show that the function which is  $f$  on  $U$  and  $g$  on  $V$  is a regular function on  $U \cup V$ . Conclude that if  $f$  is a *rational* function on  $X$ , then there is a largest open subset  $U$  of  $X$  on which  $f$  is represented by a regular function. We say that  $f$  is *defined* at the points of  $U$ .

*Proof.* Denote the function which is  $f$  on  $U$  and  $g$  on  $V$  as  $F$ . It suffices to show for all  $P \in U \cap V$  that there exists an open neighborhood  $W$  of  $P$  contained in  $U \cap V$  such that  $F$  is a rational function on  $W$ . Since  $U \cap V$  is an open subset of  $U$  and  $V$ , for all  $P \in U \cap V$  there exists open subsets  $A$  of  $U$  and  $B$  of  $V$  such that  $F$  is rational with  $h(Q), h(Q) \neq 0$  for all  $Q \in A \cap B$  and  $F = f = g$  on  $A \cap B$ , hence  $W = A \cap B \subseteq U \cap V$  is the desired open subset.

If  $f$  is a rational function on  $X$ , then the set of open subsets of  $X$  on which  $f$  is represented by regular function is nonempty, and since  $X$  is a noetherian space, there is a maximal element in this set. Then, the largest open subset  $U$  of  $X$  on which  $f$  is represented by a regular function is the union of maximal elements in this subset of open subsets.  $\square$

2. Same problem for rational maps. If  $\varphi$  is a rational map of  $X$  to  $Y$ , show there is a largest open set on which  $\varphi$  is represented by a morphism. We say the rational map is *defined* at the points of that open set.

*Proof.* Let  $\varphi$  and  $\psi$  be morphisms on open subsets  $U$  and  $V$  of a variety  $X$  to a variety  $Y$ , and suppose  $\varphi = \psi$  on  $U \cap V$ . If  $f$  is a regular function on  $Y$ , then  $\varphi^*f$  and  $\psi^*f$  are regular functions on the open subsets  $U$  and  $V$  which agree on  $U \cap V$ , so by Exercise 1 the function which is  $\varphi^*f$  on  $U$  and  $\psi^*f$  on  $V$  is a regular function on  $U \cap V$ , i.e. the map that is  $\varphi$  on  $U$  and  $\psi$  on  $V$  is indeed a rational map.

If  $\varphi : X \rightarrow Y$  is a rational map, then the set of open subsets of  $X$  on which  $\varphi$  is represented by morphism function is nonempty, and since  $X$  is a noetherian space, there is a maximal element in this set. Then, the largest open subset  $U$  of  $X$  on which  $\varphi$  is represented by a morphism is the union of maximal elements in this subset of open subsets.  $\square$

4. A variety  $Y$  is *rational* if it is birationally equivalent to  $\mathbb{P}^n$  for some  $n$  (or, equivalently by (4.5), if  $K(Y)$  is a pure transcendental extension of  $k$ ).

- (a) Any conic in  $\mathbb{P}^2$  is a rational curve.
- (b) The cuspidal cubic  $y^2 = x^3$  is a rational curve.
- (c) Let  $Y$  be the nodal cubic curve  $y^2z = x^2(x+z)$  in  $\mathbb{P}^2$ . Show that the projection  $\varphi$  from the point  $P = (0, 0, 1)$  to the line  $z = 0$  induces a birational map from  $Y$  to  $\mathbb{P}^1$ . Thus,  $Y$  is a rational curve.

*Proof.*

- (a) A conic in  $\mathbb{P}^2$  can be covered by open affine varieties that are either isomorphic to  $y = x^2$  or  $xy = 1$ . The former is isomorphic to  $\mathbb{A}^1$ , hence it is isomorphic to an open subset of  $\mathbb{P}^1$ , hence it is birationally equivalent to  $\mathbb{P}^1$ . The latter has function field isomorphic to  $k(x)$ , hence it is birationally equivalent to  $\mathbb{A}^1$ , hence it is also birationally equivalent to  $\mathbb{P}^1$ .
- (b) The cuspidal cubic has coordinate ring  $k[t^2, t^3]$ , so its function field is  $k(t)$ , hence it is birationally equivalent to  $\mathbb{A}^1$ , hence it is birationally equivalent to  $\mathbb{P}^1$ .

- (c) The line  $z = 0$  in  $\mathbb{P}^2$  corresponds to a hyperplane isomorphic to  $\mathbb{P}^1$ , so the projection  $\varphi : \mathbb{P}^2 - \{P\} \rightarrow \mathbb{P}^1$  is a morphism. In coordinates,  $\varphi$  is defined as  $(x_0, x_1, x_2) \mapsto (x_0, x_1)$ , so  $\varphi$  induces a morphism from  $Y - P$  to  $\mathbb{P}^1$ . Thus,  $\varphi(Y - P)$  is the set of all lines in  $\mathbb{A}^2$  that pass through the origin and a point in the affine nodal curve  $y^2 = x^3 + x^2$ . This is an open set in  $\mathbb{P}^1$  isomorphic to  $\mathbb{A}^1$  since it contains all lines in  $\mathbb{A}^2$  besides the one defined by  $x = \pm y$ . To further elaborate, if  $(x, y) \in \mathbb{P}^1$  with  $x \neq \pm y$  and  $x \neq 0$ , say, then write  $y = \lambda x$  with  $\lambda \neq \pm 1$ , so we have

$$\lambda^2 x^2 = x^3 + x^2 \implies x = \lambda^2 - 1.$$

So there exists a line in  $\mathbb{A}^2$  passing through a point in  $y^2 = x^3 + x^2$  with slope  $y/x$ . Rephrasing, we have shown that the map  $\varphi : Y - P \rightarrow \mathbb{P}^1$  is surjective besides at the two points  $(1, 1)$  and  $(1, -1)$ . It is also injective since the  $x_2$ -coordinate can be completely determined by the values of  $(x_0, x_1)$ , that is if  $x_0 \neq 0$ , then setting  $x_0 = 1$ , we have

$$x_1^2 x_2 = 1 + x_2 \implies x_2 = \frac{1}{x_1^2 - 1} \quad \text{or} \quad x_2 = 0,$$

and  $x_1 \neq \pm 1$  since  $x_0 \neq \pm x_1$ , and  $x_2 \neq 0$  since the only point on  $Y$  with  $x_2 = 0$  is  $P$ , and  $\varphi$  is restricted  $Y - P$ . Hence,  $\varphi$  is an isomorphism of the open subset  $Y - P$  of  $Y$  to the open subset  $\mathbb{P}^1 - \{(1, 1), (1, -1)\}$  in  $\mathbb{P}^1$ , hence  $Y$  is birationally equivalent to  $\mathbb{P}^1$ . □

5. Show that the quadric surface  $Q : xy = zw$  in  $\mathbb{P}^3$  is birational to  $\mathbb{P}^2$ , but not isomorphic to  $\mathbb{P}^2$ .

*Proof.*  $Q$  is the Segre embedding of  $\mathbb{P}^1 \times \mathbb{P}^1$  in  $\mathbb{P}^3$ , and  $\mathbb{P}^1 \times \mathbb{P}^1$  contains a copy of  $\mathbb{A}^1 \times \mathbb{A}^1$ , so  $Q$  contains a copy of  $\mathbb{A}^2$ , hence  $Q$  is birational to  $\mathbb{P}^2$ . It is an axiom of projective geometry that any two lines intersect in  $\mathbb{P}^2$ ; however, it was shown in Exercise 2.15 that there exists lines in  $Q$  that do not intersect, hence  $Q$  and  $\mathbb{P}^2$  cannot be isomorphic. □

10. Let  $Y$  be the cuspidal cubic curve  $y^2 = x^3$  in  $\mathbb{A}^2$ . Blow up the point  $O = (0, 0)$ , let  $E$  be the exceptional curve, and let  $\tilde{Y}$  be the strict transform of  $Y$ . Show that  $E$  meets  $\tilde{Y}$  in one point, and that  $\tilde{Y} \cong \mathbb{A}^1$ . In this case the morphism  $\varphi : \tilde{Y} \rightarrow Y$  is bijective and bicontinuous, but it is not an isomorphism.

*Proof.* Let  $t, u$  be homogenous coordinates for  $\mathbb{P}^1$ . Then  $X$ , the blowing-up of  $\mathbb{A}^2$  at  $O$ , is defined by the equation  $xu = ty$  inside  $\mathbb{A}^2 \times \mathbb{P}^1$ . We obtain the total inverse image of  $Y$  in  $X$  by considering the equations  $y^2 = x^3$  and  $xu = ty$  in  $\mathbb{A}^2 \times \mathbb{P}^1$ . Now  $\mathbb{P}^1$  is covered by the open sets  $t \neq 0$  and  $u \neq 0$ , which we consider separately. If  $t \neq 0$ , we can set  $t = 1$ , and use  $u$  as an affine parameter. Then we have the equations

$$y^2 = x^3, \quad y = xu$$

in  $\mathbb{A}^3$  with coordinates  $x, y, u$ . Substituting, we get  $x^2 u^2 - x^3 = 0$ , which factors. Thus, we obtain two irreducible components, one defined by  $x = 0, y = 0, u$  arbitrary, which is  $E$ , and the other defined by  $u = x$  and  $y = ux$ . This is  $\tilde{Y}$ , and  $\tilde{Y}$  meets  $E$  at the point  $u = 0$ . Similarly, if  $u \neq 0$ , then we can set  $u = 1$ , and use  $t$  as an affine parameter to obtain the equations

$$y^2 = x^3, \quad x = ty$$

in  $\mathbb{A}^3$  with coordinates  $x, y, t$ . Substituting, we get  $y^2 = t^3 y^3$ , which factors as well. Besides the exceptional curve, we have the component defined by  $1 - t^3 y = 0$  and  $x = ty$ , which does not meet  $E$ . Hence,  $E$  meets  $\tilde{Y}$  at only  $(1, 0) \in E$ . This also shows  $\tilde{Y}$  is contained in the open set defined by  $t \neq 0$ , so it is isomorphic to the affine variety in  $\mathbb{A}^3$  defined by  $u = x$  and  $y = ux$ , which is isomorphic to  $y = x^2$  in  $\mathbb{A}^2$ , hence  $\tilde{Y} \cong \mathbb{A}^1$ . □